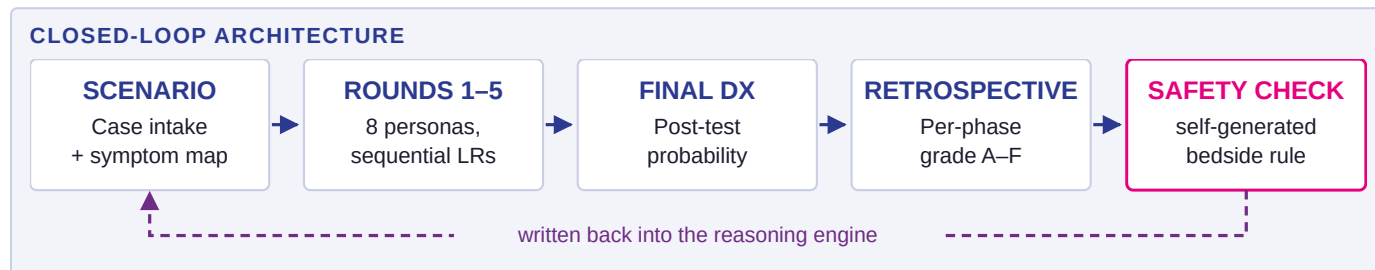


An Artificial Intelligence Clinico-Pathological Conference: Multi-Persona Bayesian Clinical Reasoning With Closed-Loop Self-Generated Bedside Safety Checks For Internal Medicine Training

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BACKGROUND & AIM

Case-based teaching rarely makes explicit **how much** each finding should move a diagnosis. We built an AI clinico-pathological conference (CPC) that reproduces the NEJM Clinical Problem-Solving format while exposing every probability revision as an auditable likelihood ratio (LR) — and that **audits itself**: the errors it makes on one case become bedside safety checks constraining the next.

METHODS

Knowledge base. 33 symptom-based chapters with LR values extracted from primary diagnostic-accuracy literature (McGee; Simel & Rennie).

Personas. Eight role-specific reasoners — attending, radiologist, pathologist, cardiologist, pulmonologist, infectious disease, neurologist and EBM specialist — activated by case-specific triggers, not run indiscriminately.

Inference. Sequential Bayesian updating on the odds scale (post-test odds = pre-test odds × LR; Fagan), retaining the full probability cascade for every competing hypothesis.

Self-audit. After the final diagnosis each phase is graded A–F against a checklist and every failure mode is rewritten as a reusable safety check; seven performance metrics are tracked across cases.

CASE 1 — PROBABILITY CASCADE

76F, six months of progressive proximal weakness, bilateral ptosis, areflexia, respiratory failure, prior SIADH.

Finding	LR+	Pre	Post
Proximal weakness + areflexia	5.0	35.0%	72.9%
Low / absent CMAPs	8.0	72.9%	95.6%
SIADH (autonomic)	3.0	95.6%	98.5%
Failed myasthenia therapy	4.0	98.5%	99.6%

Final diagnosis: Lambert-Eaton myasthenic syndrome, paraneoplastic from small-cell lung cancer — on the differential from Round 1, confirmed by Round 4.

KEY RESULT

Post-test probability can never exceed the probability mass assigned to the hypotheses actually on the list. **Bayesian reasoning is bounded by the hypothesis space** — so the teachable skill is not calibration, it is differential construction.

2 / 2

correct final Dx

100%

must-not-miss
sensitivity

6

safety checks
self-generated

CASE 2 — FIVE DIAGNOSTIC PIVOTS

38F, mail carrier, myocardial infarction before age 38, recurrent sudden chest pain, nitroglycerin non-responsive.

#	New data	Working Dx	Prob
1	Young female + prior MI	SCAD	97.5%
2	Multi-territory infarcts	Antiphospholipid	94.7%
3	Three-vessel aneurysms	Kawasaki	90%
4	Cutaneous papules	Kawasaki / NF1	50 / 10%
5	Café-au-lait + freckling	NF1 vasculopathy	>95%

Every pivot was internally coherent and confidently wrong. NF1 never appeared in the Round-1 top-ten differential, so no evidence could raise it.

SELF-GENERATED SAFETY CHECKS

Check	Fires when	Grade
Red-flag history	PMH atypical for age	D → B
Syndromic screen	Young + unclear etiology	C+ → A
Rare-cause search	Unexpected lesion pattern	B- → A
Genetic pattern	Autosomal-dominant history	B+ → A
Hypothesis-space audit	After every round	mandatory
Hypothesis humility	Always	principle

All six were derived by the system from its own Case-2 failures, not written in advance. Grade = retrospective phase grade before → after.

CONCLUSIONS

1. An AI CPC makes diagnostic reasoning **transparent and quantifiable**, with an auditable LR cascade behind every probability it states.
2. The **retrospective-to-safety-check loop** converts case failures into durable constraints, improving later performance without retraining.
3. Trigger-based multi-persona rounds preserve subspecialty depth without indiscriminate breadth.
4. Suitable for internal medicine training as a **reasoning-visible** adjunct to case conference — decision support, not decision replacement.

Metrics. LR completeness 85% / 78% · pivot efficiency Round 1 vs Round 5.

References. McGee S. *Evidence-Based Physical Diagnosis*, 4th ed. · Simel DL, Rennie D. *The Rational Clinical Examination* · Fagan TJ. *N Engl J Med* 1975;293:257.

Code and cases. github.com/htlin222/cps-skills